

Fill out your details, read the following then do this Diagnostic Test

Name (caps) _____ Student ID (9 digits)

Lab Time _____ Desk Number: Desk-_____ (eg Desk-G02-060)

How to work out what a code fragment does (and make sure you understand the construct): draw a table

The simplest and most reliable way to work out what a code fragment does (and to understand) is to draw a table. This is called tracing or dry running code, Let's work out what this fragment does:

```
int x = 9; // L1
int y = 8; // L2
if (x <= y) // L3
    {x = 2;} // L4
else
    {x = 3;} // L5
```

Executing Line L1

We write the line number and fill in the value of any variable changing, here x.

LINE	x	y	x<=y?
L1	9		

Executing Line L2

Now we move on to Line L2. Note only y changes so we only change its value in the table.

LINE	x	y	x<=y?
L1	9		
L2		8	

Executing Line L3

On Line L3 we evaluate the test (a boolean expression), Note how we fill the values in to the expression to show our working and fill in the result that is a boolean value so must be **true** or **false** as they are the two boolean values.

Writing anything else like yes or no or T or F would be wrong. Here it is false.

LINE	x	y	x<=y?
L1	9		
L2		8	
L3			9<=8? false

Executing Line L5

Because the test evaluated to false we jump to the else case so execute L5. We do not have a row at all for L4 as it is not executed. If the test had been true, then we would have had a line in the table for Line L4 but none for line L5.

LINE	x	y	x<=y?
L1	9		
L2		8	
L3			9<=8? false
L5	3		

If statement Diagnostic test: Show all your working

The results do not count for anything. This test is to allow the ECS401 team to diagnose if you need help to ensure they do not fall behind in learning to program. Understanding how assignment works is critical so if you have any problems we want to sort them out now.

Show all your working in the following. Fill out the rows in the tables to justify your answers! No correct table... no marks.

1. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
if (a == 10)  // L3
    {a = 11;} // L4
else
    {b = 11;} // L5
```

LINE	a	b	a==10?

a is now _____

b is now _____

2. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
if (a == 11)  // L3
    {a = 11;} // L4
else
    {b = 11;} // L5
```

LINE	a	b	a==11?

a is now _____

b is now _____

3. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
if (a == 11)  // L3
    {a = 12;} // L4
else
    {b = 11;} // L5
a = 13;      // L6
```

LINE	a	b	a==11?

a is now _____

b is now _____

4. What are the new values of a and b after the following code fragment has executed and what is printed?

```
String a = "5";           // L1
String b = "6";           // L2
if (a.equals("b"))        // L3
    {System.out.println(a);} // L4
else
    {System.out.println(b);} // L5
System.out.println("DONE"); // L6
```

a is now _____ **b is now** _____
Printed

LINE	a	b	a.equals("b")?	Printed

5. What are the new values of a and b after the following code fragment has executed and what is printed?

```
String a = "5";           // L1
String b = "6";           // L2
if (!(a.equals("b")))      // L3
    {System.out.println(a);} // L4
else
    {System.out.println(b);} // L5
System.out.println("DONE"); // L6
```

a is now _____ **b is now** _____
Printed

LINE	a	b	!(a.equals("b"))?	Printed

6. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5;                // L1
int b = 6;                // L2
if (a == b)               // L3
    {a = 8;}              // L4
else if (a > b)            // L5
    {a = 9;}              // L6
else                      // L7
    {a = 10;}             // L7
```

a is now _____ **b is now** _____

LINE	a	b	a==b?	a>b?

7. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;           // L1
int  b = 6;           // L2
if (a < b)             // L3
    {a = 8;}           // L4
else if (a > b)        // L5
    {a = 9;}           // L6
else                   // L7
    {a = 10;}
```

a is now _____

b is now _____

LINE	a	b	a<b?	a>b?

8. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;           // L1
int  b = 6;           // L2
if (a >= b)            // L3
    {a = 8;}           // L4
else if (a < b)        // L5
    {a = 9;}           // L6
else                   // L7
    {a = 10;}
```

a is now _____

b is now _____

LINE	a	b	a>=b?	a<b?